BREAK INTO AEROSPACE

Applied Aerodynamics & CFD

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Applied aerodynamics and CFD engineering is the discipline that determines what shape an aircraft takes, how much fuel it burns, and how it behaves at the edges of its flight envelope. It is one of the oldest formal disciplines in aerospace and one of the most computationally intensive: modern simulations run on tens of thousands of processor cores for days.

The field spans four interconnected domains: ***Low-speed aerodynamics*** *(takeoff/landing),* ***Transonic aerodynamics*** *(cruise),* ***Turbulence modelling*** *(the analytical bedrock), and* ***Shape optimisation*** *(closing the loop). Engineers in demand work across legacy OEMs (Airbus, Boeing), research institutions (NASA, DLR, ONERA), motorsport (Formula 1), and the growing eVTOL/UAM sector. This guide targets the physical intuition and numerical rigour required to bridge the gap between "CFD operator" and true aerodynamicist.*

PART I · WHAT APPLIED AERODYNAMICS & CFD ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — Employers screen for physical intuition and mathematical fluency with governing equations rather than software competence. *Distinguishing Question:* "My RANS simulation shows a shock at 55% chord, but the experiment puts it at 48%. What hypotheses explain this?" A strong candidate reasons through transition location, turbulence models, and geometry fidelity rather than just clicking buttons.

- **Mid-Level (Engineer, Senior Engineer, ∼3–8 years)** — Interviews become problem-specific. The interviewer assesses your ability to design CFD campaigns for specific design questions, interpret experimental results that disagree with simulations, and guide shape optimisation toward physically meaningful results. *Current Context:* Major hiring is driven by the A321XLR fairing work, 777X high-aspect-ratio wing aeroelasticity, and the CFM RISE open fan installation aerodynamics.

- **Senior (Principal Engineer, Technical Lead, Chief Aerodynamicist)** — Employers look for engineers who define the aerodynamic development strategy. You must be able to stand in front of a Chief Engineer and explain why a wing needs a redesign 18 months before freeze, backed by computational evidence and physical reasoning. The flight test correlation credential is the gold standard here.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
Boundary Layer Theory	Understanding transition and separation quantitatively. Knowing that a separated leading edge equals stall and being able to diagnose this in RANS or experiments.
Transonic Aerodynamics	Mastery of shock physics, wave drag, and the supercritical aerofoil. Explaining shock-boundary layer interactions and the buffet boundary on a cruise polar.
RANS Turbulence Modelling	Knowing not just how to run code, but when to select k-ω SST over Spalart-Allmaras for adverse pressure gradients, and recognizing when RANS is fundamentally insufficient.
Large Eddy Simulation (LES)	Understanding the zonal approach (RANS for attached flow, LES/DES for separation). Knowing the computational cost ratios and mesh transition requirements.
Wind Tunnel Testing	Competence in wall corrections, sting effects, and wake rake data reduction. The ability to critically compare CFD predictions against tunnel results is a core validation skill.
Aerofoil Design & C_p Analysis	Reading a C_p plot to identify laminar separation bubbles, turbulent reattachment, and buffet margin. Tools like XFOIL are considered baseline literacy.
Drag Accounting	Decomposing total drag (friction, induced, wave, interference). The ability to reason in "drag counts" (1 count $= 0.0001$ in C_D) for commercial configurations.
Adjoint Shape Optimisation	Computing sensitivities of an objective function to design variables. Understanding multi-point optimisation and manufacturing constraints in the design loop.

How Aerodynamics Interviews Differ from Generic Roles

- **Physical interpretation precedes the numerical result.** The interview does not end with $C_D = 0.0247$. It ends when you explain if that number makes physical sense relative to wetted area and form factor.

- **Validation is the engineering product.** CFD without validation is just a colorful picture. Probing mesh sensitivity and comparing against experimental benchmarks (like the NASA Common Research Model) is the standard for reliable evidence.

- **The RANS-limit is a core competency.** You must know when RANS fails (massively separated flows, noise, dynamic stall). Methodological overconfidence is a major red flag in professional aerodynamics groups.

- **Multidisciplinary interfaces are always present.** An aerodynamic gain that breaks the structural load limit is a failure. You must understand how your outputs feed structural loads and performance certification.

Part II · 20 Specific Preparation Tasks

Intern / Graduate

6–8 h

1. Derive the Navier-Stokes equations and explain the RANS averaging process **Skill:** The governing equations at the depth practitioners must understand. **Objective:** Derive the equations from the Reynolds transport theorem. Explain the Reynolds stress tensor, the turbulence closure problem, and the Boussinesq hypothesis. **Done Signal:** You can derive the incompressible RANS equations, identify the stress tensor physically, and explain what k-ω SST adds to the Spalart-Allmaras model.

5–6 h

2. Understand the boundary layer — transition, separation, and their engineering consequences **Skill:** Viscous flow physics as a design constraint. **Objective:** Use the von Kármán integral momentum equation. Understand Thwaites' criterion for separation and the Hartree parameter. **Done Signal:** You can estimate momentum thickness growth for a flat plate and explain the physical meaning of shape factor H for laminar vs. turbulent states.

5–6 h

3. Master transonic shock physics — the normal shock and isentropic flow relations **Skill:** Compressible flow physics for cruise aerodynamics. **Objective:** Apply Rankine-Hugoniot conditions and Prandtl-Meyer expansion logic. Interpret C_p distributions to find shock strength. **Done Signal:** You can compute Mach jumps/pressure ratios without tables and explain the Korn equation's relationship to drag divergence.

2–3 h

4. Prepare behavioural answers connecting computational/experimental results to decisions **Skill:** Validation-surprise emphasis. **Objective:** Focus on a time simulation and experiment disagreed. How did you investigate? **Done Signal:** You have stories ready that include specific drag counts or C_p peak discrepancies and your diagnostic process (e.g., mesh refinement or tunnel wall interference).

5–7 h

5. Build a complete aerodynamic analysis of a real aerofoil using XFOIL **Skill:** Rapid aerofoil assessment. **Objective:** Generate lift/drag polars for NACA and supercritical sections. Observe bubble-and-reattachment kinks at low Reynolds numbers. **Done Signal:** You have computed C_l-α slopes and identified transition locations from C_f distributions for at least two sections.

10–15 h

6. Get hands-on with a 3D CFD solver on a real geometry (NASA CRM) **Skill:** End-to-end CFD workflow competence. **Objective:** Geometry import to post-processing. Compare results against NTF experimental data. **Done Signal:** You can describe your mesh approach (y^+ achieved, cell count), convergence criteria, and sources of discrepancy with reference data.

Mid-Level *(∼3–8 years)*

All graduate tasks still apply. Stories must come from professional contexts involving real aerodynamic design decisions and traceable engineering consequences.

5–6 h

7. Build thorough working knowledge of turbulence model selection **Skill:** The judgment that determines CFD reliability. **Objective:** Differentiate between SA, k-ω SST, and hybrid methods (DES/IDDES). Understand separation onset physics. **Done Signal:** You can explain why k-ω SST outperforms k-ϵ in adverse pressure gradients and state the cost ratios between RANS and LES.

4–5 h

8. Understand drag prediction uncertainty and the AIAA Drag Prediction Workshop findings **Skill:** Quantitative performance assessment. **Objective:** Study DPW scatter for the DLR-F6/CRM. Learn which features cause inter-code disagreement (wake drag, junctions). **Done Signal:** You can describe the typical scatter in RANS drag predictions (∼15–25 counts) and report a result with proper uncertainty bounds.

2 h

9. Prepare a technically specific story about a CFD-experiment discrepancy **Skill:** Validation engineering competence. **Objective:** Focus on a specific diagnostic (e.g., Mokry's method for wall interference or shock mesh refinement). **Done Signal:** Your story identifies the configuration, reference data, the correction magnitude in drag counts, and the residual uncertainty.

4–5 h

10. Understand wind tunnel testing methodology and the test-to-flight scaling challenge **Skill:** Experimental aerodynamics competence. **Objective:** Learn 6-component balance resolving, wake rake drag surveying, and Reynolds number scaling issues. **Done Signal:** You can describe how sting interference is corrected and why cryogenic tunnels are used for Reynolds number matching.

5–6 h

11. Understand adjoint-based shape optimisation — sensitivities and constraints **Skill:** High-dimensional design closure. **Objective:** Explain how adjoint equations are derived and why cost is independent of design variable count. **Done Signal:** You can physically describe what a surface normal sensitivity represents and state the limits of gradient-based methods.

4–5 h

12. Build working knowledge of high-lift aerodynamics and maximum lift prediction **Skill:** Aerodynamics of the approach/departure margins. **Objective:** Learn multi-element flow physics (confluent boundary layers). Understand why RANS over-predicts C_{Lmax}. **Done Signal:** You can explain the three mechanisms by which a slat increases lift and how stall character affects handling quality certification.

5–6 h

13. Prepare for an aerodynamic analysis problem under interview conditions **Skill:** Analytical speed and structure. **Objective:** Practice C_p interpretation, first-principles drag estimation, and CFD troubleshooting scenarios. **Done Signal:** You have completed three practice problems in under 20 minutes each, explaining the physical significance of your results.

Senior Level *(Principal Aerodynamicist, Technical Lead, Chief)*

4–6 h

14. Prepare to walk through the complete aerodynamic development of a component **Skill:** End-to-end development ownership. **Objective:** Trace a component from initial shape and optimisation to tunnel validation and flight correlation. **Done Signal:** You can quantify the improvement (e.g., drag counts recovered) and identify the specific tunnel result that drove a redesign.

4–5 h

15. Prepare a technically grounded position on one active aerodynamics frontier **Skill:** Forward-looking engineering depth. **Objective:** Deep dive into natural laminar flow (crossflow instability), open fan pylon interactions, or BWB high-lift challenges. **Done Signal:** You have a 3-minute technical talk identifying a **specific unsolved engineering problem** on your chosen frontier.

4–5 h

16. Prepare to lead a technical discussion on an aerodynamic design trade-off **Skill:** Architectural speed and reasoning. **Objective:** Reason through aspect ratio vs. sweep vs. wing weight fuel burn penalties live. **Done Signal:** You can arrive at a defensible recommendation for a narrowbody fleet targeting 15% fuel reduction, stating explicit trade-offs.

4–5 h

17. Understand the aerodynamics-structures aeroelastic coupling **Skill:** Multidisciplinary high-aspect-ratio wing design. **Objective:** Explain wash-out under positive bending and how jig twist is determined to hit cruise targets. **Done Signal:** You can quantify the drag counts difference between rigid and deformed shape CFD and describe validation using flight measurements.

4–5 h

18. Build quantitative estimation fluency for aerodynamic performance — without simulation **Skill:** Senior-speed sanity checking. **Objective:** Use Prandtl-Schlichting, Oswald efficiency, and Korn equations for order-of-magnitude checks. **Done Signal:** You have worked through five estimation problems (e.g., cruise C_L from weight) stating assumptions and uncertainty out loud.

All Levels

1–2 h

19. Prepare a specific answer to "Why applied aerodynamics?" reflecting genuine engagement
Skill: Intellectual character signal. **Objective:** Name an active challenge (e.g., crossflow instability or eVTOL hover-to-forward transition) that excites you. **Done Signal:** You have a 90-second answer that proves you are a specialist, not just a generic fluid mechanics applicant.

1–2 h

20. Map your background to a specific employer type, domain, and tool set **Skill:** Precise self-positioning. **Objective:** Understand where you sit on the spectrum from research (LES/hypersonics) to production (overnight RANS iterations). **Done Signal:** You can state your targeted role, tool fluency (elsA, Fluent, etc.), and experimental exposure in under 60 seconds.

Summary Reference Table

#	Task Description	Target Level	Effort
1	Derive RANS equations — closure problem, turbulence models	All	6–8 h
2	Boundary layer theory — transition, separation, Thwaites	All	5–6 h
3	Transonic shock physics — relations, Korn, C_p interpretation	All	5–6 h
4	STAR stories — CFD/experiment, optimisation, own error	All	2–3 h
5	XFOIL aerofoil analysis — polars and C_p distributions	All	5–7 h
6	3D CFD on benchmark geometry — validation workflow	All	10–15 h
7	Turbulence model selection — RANS vs LES vs DES	Mid+	5–6 h
8	Drag prediction uncertainty — scatter and DPW findings	Mid+	4–5 h
9	CFD-experiment discrepancy diagnosis story	Mid+	2 h
10	Wind tunnel methodology — balance and scaling	Mid+	4–5 h
11	Adjoint-based shape optimisation sensitivities	Mid+	5–6 h
12	High-lift aerodynamics — slat/flap flow physics	Mid+	4–5 h
13	Timed aerodynamic problem-solving	Mid+	5–6 h
14	Complete development story — tunnel to flight correlation	Senior	4–6 h
15	Active frontier position (Laminar, Open Fan, BWB)	Senior	4–5 h
16	Aerodynamic design trade-off discussion (live)	Senior	4–5 h
17	Aeroelastic coupling — wash-out and jig twist	Senior	4–5 h
18	Quantitative estimation fluency — without CFD	Senior	4–5 h
19	Technically specific "Why aerodynamics?" answer	All	1–2 h
20	Employer and tool set self-positioning	All	1–2 h

The Structural Reality

Applied aerodynamics and CFD is the engineering discipline where the most expensive computers in the world are used to study a phenomenon — turbulent flow — that remains one of the most poorly understood in all of physics. The Navier-Stokes equations have been known for 180 years, but the turbulence problem they produce at high Reynolds numbers remains partially unsolved.

Every aerodynamicist who claims a CFD result is reliable without demonstrating validation and sensitivity analysis is producing an output that is potentially dangerously wrong. The community has learned this through costly failures in shock prediction, aeroelastic coupling, and C_{Lmax} over-prediction.

Successful engineers hold computational capability and physical intuition simultaneously. They know what a C_p plot for a supercritical wing should look like before the code runs. They pre-visualise the flow and use simulations to confirm or contradict expectation. This ability to "read" a flow cannot be faked; it is the core of what technical interviews assess.

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